

Unit 406 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 35.1 ac, BAF 20 variable-radius cruise on a 300 ft grid, 17 plots, 205 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	241 sq ft/ac (SE 2.3 %)	Type	Sierran mixed conifer
Trees per acre	71	CWHR size / density	5 / D (cover 93 %)
QMD	24.9 in	SDI	297 (50 % of max 598)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	241 sq ft/ac 24.9 in	BA QMD	0 sq ft/ac 0.0 in
TPA	71	TPA	0
Volume	56.8 MBF/ac (10335 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~170 sq ft/ac BA	Sampling error	4.9 % at 95 % (t = 2.12, n = 17)
Species (BA)	WF 82, PP 60, IC 42, DF 34, SP 22	Stratum standard	40 % - MEETS; 17 plots vs 20-plot practice: SHORT by 3

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	1	1.6	1.2	32	0.2
12-15.9	12	12.2	14.1	432	2.4
16-19.9	17	11.5	20.0	688	3.8
20-23.9	41	18.1	48.2	1773	9.8
24-27.9	29	9.6	34.1	1376	7.6
28-31.9	34	8.2	40.0	1727	9.5
32-35.9	21	4.0	24.7	1153	6.3
36-39.9	19	2.9	22.4	1106	6.1
40-43.9	13	1.6	15.3	813	4.5
44-47.9	7	0.7	8.2	468	2.6
48-51.9	5	0.4	5.9	336	1.9
52-55.9	1	0.1	1.2	80	0.4
56-59.9	3	0.2	3.5	211	1.2
60-63.9	2	0.1	2.4	140	0.8
All	205	71.2	241.2	10335	56.8

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	70	21.1	82.4	26.7	3730	20.5
PP	51	21.5	60.0	22.6	2396	13.2
IC	36	12.1	42.4	25.4	1856	10.2
DF	29	10.2	34.1	24.8	1395	7.7
SP	19	6.3	22.4	25.5	958	5.3
All	205	71.2	241.2	24.9	10335	56.8

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 406: plots on the 300 ft grid (orange = QA flag)

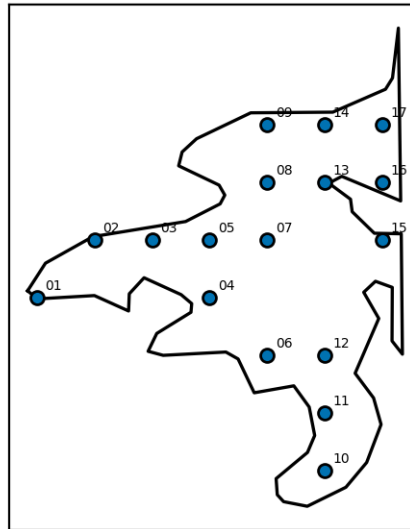


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted; 3 more plots would meet the 20-plot practice

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
406-01	13	260	186	ok
406-02	12	240	171	ok
406-03	12	240	259	ok
406-04	13	260	212	ok
406-05	11	220	177	ok
406-06	14	280	114	ok
406-07	12	240	260	ok
406-08	11	220	206	ok
406-09	10	200	185	ok
406-10	11	220	153	ok
406-11	12	240	212	ok
406-12	11	220	97	ok
406-13	12	240	147	ok
406-14	13	260	170	ok
406-15	11	220	166	ok
406-16	13	260	139	ok
406-17	14	280	152	ok